

Notes on the proposal: connecting papers, data, and the search for the fundamental tension

Companion notes to `proposal_eee_draft.tex`

What these notes are. Suggestions to organize the ideas already in `proposal_eee_draft.tex` based on (i) the five papers in `papers/` and (ii) the empirical evidence already in `context/`. The goal is to help you find (a) the fundamental tension that makes the paper revealing, (b) the CF that exploits that tension, and (c) concrete answers to the four modelling questions you flagged: unobserved heterogeneity, learning with discrete signals, the school’s role in a Stackelberg game, and equilibrium.

1 The fundamental tension and what it should buy you

Where the proposal currently sits. The current draft mixes two ideas: (1) students learn about ability from grades and recommendations, (2) schools may distort recommendations strategically. Each is interesting, but neither alone is a top-5 contribution: (1) is well-mapped by AAMR (2025) and Pastorino (2024), and (2) without a sharp identification argument is indistinguishable from descriptive school heterogeneity.

The tension that the data already gives you. The `update_april_2026` memo identifies it without naming it: *the incentive of a school to use its informational advantage strategically depends on whether the school keeps the student*. Cycle-1-only schools (41% of the sample) lose the student mechanically at the end of grade 2. Cycle-1+2+3 schools keep the student and face capacity constraints, pass-rate concerns, and track-composition incentives. Within the same students-types pool, the recommendation behaviour of these two school architectures should differ *only because of incentives*. This is the Pastorino (2024) “differential informativeness of jobs” argument relabeled for schools, and it is the cleanest identification source you have.

Sharpening it. In `update_08_12` the descriptive evidence shows that, conditional on standardized exam grades, IQ, and effort, teachers still rely heavily on *relative class rank* when issuing horizontal (STEM/non-STEM) advice. Relative rank is exactly the object that a strategic school manipulates if it is capacity-constrained. So the tension is not just “schools may be biased”: it is that *the very statistic teachers use (class rank) is endogenous to the school’s allocation problem*, and this distorts the signal value of the recommendation for the student’s posterior about her own track-specific ability.

The revealing CF. Once that tension is the engine of the model, the natural CF is not “ban tracking” (Fu–Mehta) or “move test cutoffs” (Larroucau–Rios). It is:

1. **Information policy:** replace the school-issued recommendation with a centrally-issued, capacity-free signal of equal informativeness about θ (e.g., a standardized tracking score). What share of mismatch (downstream switching/dropout/HE field of choice) is due to schools’ strategic distortion, vs. to genuine informational frictions in the student’s own learning?
2. **Decoupling supply from advice:** hold school capacities fixed but require schools to issue advice as if they had no retention incentive (the “cycle-1 benchmark”). This isolates the welfare cost of the Stackelberg leadership of the school.
3. **Stackes-of-information rebalancing:** make recommendations binding (as B/C certificates already are) vs. purely advisory, and study who gains and who loses by SES, ability and gender. This connects to Larroucau–Rios’s mechanism-design CFs in a way that uses your strategic sender as the policy lever rather than the application algorithm.

Of these, (1) is the *revealing* one because it directly decomposes the welfare effect of recommendations into “information” and “strategic distortion” components. The other two are robustness/design exercises.

Why this would be top-5. It is the first paper that estimates a Bayesian-persuasion-style information design model in a real schooling setting, with continuous and discrete signals jointly, with a credible identification of the strategic margin (cycle-length variation), and where the counterfactual answers a question that none of the related papers can: what is the welfare value of *the strategic component* of teacher advice? Fu–Mehta quantify the value of tracking, Pastorino quantifies the value of learning, AAMR quantifies the value of imperfect information; you would quantify the value of *strategic* information provision in education.

2 Operationalizing unobserved heterogeneity (θ^k, θ^u)

What you can identify from your measurement system. Your data is unusually rich. Following AAMR (2025) and Cunha–Heckman–Schennach (2010), a factor model in baseline measurements pins down the joint distribution of the *initially-known* types θ^k :

- **Cognitive:** Verbal IQ, Numeric IQ, Spatial IQ at start grade 1 $\Rightarrow$ at least two factors ($\theta_{\text{num}}^k, \theta_{\text{verb}}^k$) identified from intra-test correlations.
- **Non-cognitive:** performance motivation, test anxiety, conscientiousness, agreeableness, extraversion (Big-5 already estimable from the well-being/self-concept battery) $\Rightarrow$ a non-cog factor θ_{noncog}^k (your update_08_12 already shows conscientiousness is independently predictive of advice).
- **Parental press / SES:** the parental questionnaire (education, occupational situation, educational climate) $\Rightarrow$ a fourth factor $\theta_{\text{parental}}^k$ that the descriptive evidence shows matters for sorting at grade 1 but is largely absorbed by school FE for advice — i.e., its main role is in the *first-stage* school choice, not in recommendation.

Three points that follow:

1. Your draft says “ θ^k has 4 elements”. The data supports 4–5: numeric, verbal, non-cog, parental press, and arguably a separate motivation/anxiety dimension. Test the rank of the measurement system before fixing the dimension.
2. Initial sorting $F(\theta^k | s)$ is non-trivial (update_april.2026). You have two clean handles for it: (a) recover θ^k from baseline measurements, condition on it; (b) use distance-to-school as an excluded shifter in a joint first-stage school choice model. Your memo proposes both. I would do (a) for the working paper and add (b) as a robustness exercise: distance is exogenous to the recommendation rule conditional on θ^k , but residential location is endogenous in the long run, so combining the two makes the argument airtight.
3. **What is θ^u here?** In AAMR/Pastorino it is fundamental ability. In your setting the natural object is *track-specific match quality*: the realization the student would experience if assigned to ASO vs. TSO vs. BSO, and within ASO/TSO to STEM vs. non-STEM. The student has a prior over the vector of match qualities; grades update beliefs about each component (because cognitive skills predict track-specific outcomes differently); recommendations add discrete information. This is the AAMR multivariate correlated-learning structure applied to your tracks.

Practical recipe. Build the measurement system at $t = 1$ to recover $F(\theta^k)$ via mixtures-of-normals (Bunting–Diegert–Maurel 2025 allows you to do this without elicited beliefs and without assuming the measurements are selection-free, which is convenient because your tests *are* given to all students). Use a low number of discrete types $I = 3$ or $I = 4$ (Pastorino) for tractability if a continuous factor model is too expensive in the dynamic estimation. You then treat θ^k as known to agent and school but unknown to the econometrician, and let the unknown θ^u enter only through track-specific match quality, where learning operates.

3 Modelling learning with discrete signals

Continuous + discrete signal structure. At each end-of-year $t \in \{1, 2, 4, 6\}$ you observe two continuous signals (Math and Dutch test scores) and two discrete signals (the teacher’s choice advice, and at grades 2/4/6 the A/B/C certificate). Following AAMR/BDM, write

$$G_{ijt} = \mu_j(\theta_i^k, X_{it}) + \lambda_j \theta_{ij}^u + \varepsilon_{ijt}, \quad j \in \{\text{Math, Dutch}\}, \quad \theta_i^u \sim \mathcal{N}(0, \Sigma_u),$$

with Σ_u allowed to be non-diagonal (correlated learning across track dimensions, as in AAMR). Posterior on θ^u is Gaussian conjugate.

Adding the discrete recommendation. The recommendation $R_{it} \in \mathcal{R}$ that the school issues is informative about θ^u *only* insofar as the school uses additional information (its own observations of effort, peer comparisons, knowledge of the cohort distribution). Two options for the student’s updating, in increasing order of complexity:

- **Reduced-form signal extraction.** Treat the recommendation as a noisy categorical signal of θ^u with school-type-specific informativeness α_s, β_s (Pastorino’s α_{jk}, β_{jk}). The student’s posterior depends on the school-type-specific likelihood $\Pr(R_{it} = r \mid \theta^u, s)$. Since the student observes s , she knows the likelihood she should use. *This is where strategic scepticism enters:* a Bayesian student facing a school she suspects is strategic uses a flatter (less informative) likelihood. Heterogeneity in trust can be modelled as random coefficient in α_s .
- **Structural signal extraction with strategic school.** The student knows the school’s recommendation rule $R_{it} = \rho_s(\theta^k, G_{it}, \text{cohort rank, capacity})$ and inverts it. Then R_{it} is informative about θ^u both directly (because the school sees more) and indirectly (because the school’s own decision rule reflects what the school’s information was). This is the Bayesian-persuasion / strategic-sender layer.

The first is enough for a first paper. The second is the version that delivers the “strategic information design” contribution. I would estimate both and use the difference as the main quantitative result.

Identification of the discrete-signal channel. Critical observation: in your data there are cases where Math grade and Dutch grade disagree (high Math, low Dutch or vice versa) and the recommendation sometimes goes one way, sometimes the other. This within-student variation in the alignment of continuous signals identifies how the school weights each signal and, residually, what *additional* information the school is using (effort, behaviour, capacity, family pressure). The `update_08.12` finding that “even after conditioning on IQ, std grades and effort, teachers still use relative class rank” is directly this residual: it isolates the school’s private information component, which is the input to the strategic distortion.

Pastorino-style update formula. The expression in your draft,

$$P^s \mid \theta = \frac{p(\theta) p(r \mid \theta)}{\sum_{r \notin R} p(r \mid \theta)},$$

should be rewritten as a Bayesian update of the joint posterior on $(\theta_{\text{ASO}}^u, \theta_{\text{TSO}}^u, \theta_{\text{BSO}}^u, \theta_{\text{STEM}}^u, \theta_{\text{nSTEM}}^u)$:

$$f(\theta^u \mid G_t, R_t, s) \propto f(\theta^u \mid G_t) \cdot \Pr(R_t \mid \theta^u, s),$$

with the first factor coming from the Gaussian update on grades and the second the discrete-signal likelihood. The school index s enters the second factor because the recommendation rule is school-type-specific. This makes explicit that grades and recommendation are *correlated* signals, and the identification of Σ_u comes from the joint distribution of grades and recommendations across years.

4 The school’s problem and the Stackelberg game

Schools as Fu–Mehta with information design. Fu–Mehta (2017) gives you the right scaffolding: schools choose a tracking regime (in your case, recommendation policy) and effort, parents (or students) respond. The piece you add is that in your setting the school’s lever is not (only) effort or assignment but the *informativeness* of a discrete signal that students use to update beliefs.

A workable specification:

- **State of the school:** cohort distribution of types $G_s(\theta^k)$ inherited from sorting at grade 1; capacity constraints $\bar{n}_{\text{ASO-stem}}, \bar{n}_{\text{ASO-nstem}}, \dots$ (cycle-length and supply-driven — you observe which subtracks each school offers).
- **Action:** a recommendation rule $\rho_s : (\theta^k, G_t, \text{rank}, \text{effort}) \mapsto \Delta(\mathcal{R})$ — a (possibly mixed) mapping from observables to a probability distribution over recommendations $\mathcal{R} = \{\text{ASO-stem}, \text{ASO-nstem}, \text{TSO-stem}, \text{TSO-nstem}, \text{BSO}, \dots\}$.
- **Objective:** weighted sum of (a) student average final outcome (test scores at grade 6, HE entry, labour-market outcome), (b) share of students above proficiency (the A-certificate share is observable in your data and is the natural Fu–Mehta proficiency target), (c) cost of having to repeat / over-stretch capacity. Cycle-1-only schools put weight ≈ 0 on (b) and (c) for grades ≥ 3 because the student leaves — this is the source of identification.
- **Constraint:** capacity constraints; centrally-set rules for who can be assigned to which track (B/C certificate rules).

Timing (Stackelberg).

- Stage 1. School commits to recommendation rule ρ_s (chosen once per cohort, plausibly stable across cohorts within a school — testable using the multiple cohorts in LOSO).
- Stage 2. Students observe school type s and choose school at grade 1, knowing distance, peers, ρ_s (or only the school type, if you take the empirical finding that A-certificate rates do not predict choice as evidence of myopic enrollment — this is the simpler benchmark).
- Stage 3. Period-by-period: student exerts effort, receives grade, school observes everything and issues recommendation according to ρ_s , student updates beliefs.
- Stage 4. At grade 2 (the binding moment): school issues certificate + recommendation; student chooses track conditional on it.
- Stage 5. Continue dynamically until grade 6 / HE entry / LM entry.

Equilibrium. An equilibrium is a tuple $(\rho_s^*)_{s \in \mathcal{S}}, \pi^*$ where:

1. For each school type s , ρ_s^* maximizes the school’s objective given the induced student best-response π^* .
2. For each student type, $\pi^*(\theta^k, s, G_t, R_t)$ is the Bayesian-Nash dynamic discrete choice over tracks and effort, given posterior beliefs over θ^u that are computed using ρ_s^* as the likelihood of the recommendation signal.
3. The cohort-level distribution of types and outcomes implied by (ρ_s^*, π^*) is consistent with the capacity constraints and proficiency objective fed back into the school’s problem.

This is a perfect Bayesian equilibrium of an information-design game with a population of receivers, in the spirit of Kamenica–Gentzkow with heterogeneous priors and an outside option.

Static vs. dynamic. Your draft asks “static or dynamic”. My recommendation: *static for the school, dynamic for the student*. A school’s recommendation rule is plausibly committed for the cohort and stable across cohorts (you can test this with cross-cohort variation). A student is genuinely forward-looking: choosing ASO-stem at grade 3 has option value over HE STEM. So model the student as a finite-horizon dynamic discrete choice agent (AAMR-style) and the school as a one-shot designer of the recommendation rule. This is much more tractable than a fully dynamic school problem and probably loses very little.

5 Identification: where each ingredient comes from

- $F(\theta^k)$ and $F(\theta^k \mid s)$: factor model on baseline measurements (BDM 2025, Cunha–Heckman–Schennach 2010). Initial sorting is a separate object, identified using distance.
- **Grade equation parameters and Σ_u** : BDM/AAMR semiparametric ID from the panel of grades and choices. You do not need elicited beliefs because BDM (2025) shows the model is identified without them under normality. This is the strongest part of your data.
- **Recommendation rule ρ_s** : reduced-form multinomial logit on observables, with school-type fixed effects, gives you the descriptive ρ_s . The structural piece needs the equilibrium condition that the school is optimizing given student best-response. The cycle-1 vs. cycle-1+2+3 contrast plays the role of an excluded variation that shifts the school’s incentives but not the student’s primitives.
- **Student trust / strategic scepticism**: identified from the gap between how strongly students react to a recommendation in cycle-1 schools (where the school is presumed truthful) vs. in cycle-1+2+3 schools (where the school is suspected of distortion), conditional on θ^k . Heterogeneity in trust by SES is identified from the same source.
- **Welfare and CFs**: once ρ_s^* is identified for each school type, simulate the student dynamic problem under (a) the actual ρ_s^* , (b) the cycle-1 (presumed truthful) ρ_s^* applied uniformly, (c) a centrally-set, common, capacity-free signal of the same Blackwell informativeness as the actual recommendations, and compute outcome distributions.

6 Concrete edits I would suggest to the proposal

1. Frame the contribution as “estimating a strategic-information design model in education” rather than “adding a learning model with schools”. Cite Kamenica–Gentzkow, Bergemann–Morris, and frame schools as senders in the spirit of Pastorino’s two-sided learning.
2. Replace the open “static vs. dynamic” question by the explicit static-school / dynamic-student split.
3. Make the cycle-1 vs. cycle-1+2+3 contrast the central identification source for the strategic margin, and announce it in the introduction. This is what makes the paper feel “revealing” rather than descriptive.
4. Promote the multivariate correlated-learning structure (one θ^u per track-subtrack combination) explicitly. AAMR is the template; your tracks are 5, theirs are also 5.
5. Drop peer effects from the model. They are interesting but the capacity-constraint argument already captures the main supply-side force, and adding peer effects on top will create an endogeneity/identification problem that is not central.
6. Drop effort as a separate model object on the student side. As you rightly note in the draft, it is captured in learning. Keep effort only as an input the school observes (it is in your data) so that the recommendation rule can depend on it. This matches `update_08_12` which finds effort matters for vertical (A/B/C) but not horizontal advice.
7. In the CF section, lead with the centrally-issued signal (CF 1 above) as the main result, and use the other CFs as robustness.

7 Risks and open questions

- **Are cycle-1 schools really non-strategic?** They have no retention incentive but they may still care about reputation (placement of their students into good cycle-2 schools). Test by looking at whether cycle-1 schools whose graduates feed into specific cycle-2 schools have systematically different recommendations.

- **Identification of θ^u vs. time-varying shocks.** AAMR/BDM both impose normality. Verify that mixture-of-normals fits the empirical distribution of grades by track at $t = 1$ before locking in.
- **Stability of ρ_s across cohorts.** The model's Stackelberg structure assumes school commits. Test using LOSO's multi-cohort follow-ups whether ρ_s moves with cohort composition. If it does, you may need a school problem with state rather than a one-shot design.